

EXA-GP Integration - Position & Approach

Piter Garcia (RIT Student) <pzg8794@rit.edu>
 Reply-To: pzg8794@rit.edu
 To: Daniel Krutz <dxkvse@rit.edu>

Wed, Dec 10, 2025 at 5:55 PM

Hi Dan,

Thank you for sharing that paper again—I agree, it's highly relevant to our work, which is why I have kept it open to reference it as we wrap up this paper.

Here are the modified responses to your two requests:

Specific Problem Addressed in Path Selection:

The main challenge is to dynamically select multi-hop entanglement routes that maintain high end-to-end fidelity in noisy, lossy, and resource-limited quantum networks. The current gap is that most work assumes known, stationary link success rates, ignoring adversarial attackers—assumptions that break down in deployed quantum networks. Our approach targets this gap by handling both stochastic uncertainty and adversarial threats while simultaneously managing constrained qubit allocation across path links.

ML Aspect Designed to Solve Unique Quantum Problem:

We employ Adversarial Group Neural Bandits, combining neural networks with multi-armed bandit decision-making. We treat each candidate path as a group and each qubit allocation as an arm to learn the nonlinear relationship between allocation and entanglement success probability, while UCB-based confidence bounds enable robust exploration—exploitation under stochastic link failures and adversarial attacks, achieving provably sublinear regret without prior knowledge of link success rates.

Best,
 Piter

On Wed, Dec 10, 2025 at 5:47 AM Daniel Krutz <dxkvse@rit.edu> wrote:

Also, sorry if I already shared this with you, but this paper could be of some interest:

IEEE Xplore Full-Text PDF:
 ieeexplore.ieee.org



On Dec 9, 2025, at 2:42 PM, Daniel Krutz <dxkvse@rit.edu> wrote:

Hi Piter, could you answer the following with 2-3 sentences for each area.

Specific Problem Addressed In Path Selection: <— Here is what you said before, could you mention a few words about the current gap in the body of knowledge that will be addressed as well?

" The main challenge is to dynamically select multi-hop entanglement routes that maintain high end-to-end fidelity in noisy, lossy, and resource-limited quantum networks. At the same time, the routing strategy must adapt online to changing channel conditions and potential attacks while balancing fidelity, purification overhead, and coherence lifetimes."

ML Aspect Designed to Solve Unique Quantum Problem: <— Mention the specific ML that we will be using

Thanks
 -Dan

On Dec 1, 2025, at 9:13 AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hey Dan,

Here are the answers to the questions we discussed in our previous meeting:

Does the EXPNeuralUCB paper do path exploration?

Yes. It treats each candidate path as a group and uses an adversarial bandit (EXP3-style) over paths, so

it actively explores and updates beliefs over paths while simultaneously using a NeuralUCB learner to optimize qubit allocations within each selected path.

Should we compare path exploration strategies?

Yes—it's worth explicitly contrasting our adversarial, reward-driven path exploration with stochastic best-path methods like BeQuP, LinkSelFiE, and QPSEL, which assume mostly stochastic feedback and aim to identify a single highest-fidelity route under resource constraints. In addition, comparing against less optimal baselines—including the original EXPNeuralUCB setup and the routing strategies it reported improvements over—lets us show both how our approach improves on our own earlier design and how it behaves relative to newer quantum-routing algorithms.

Should we borrow from the new path-exploration paper and/or next steps?

Yes. That's essentially how this project has been moving forward: we've already borrowed and recombined strategies from recent bandit and quantum-routing work (adversarial NeuralUCB, pursuit-style contextual bandits, stochastic forecasting, dynamic capacity and allocator variants) to build our own hybrid models, which in our evaluations significantly outperformed the original EXPNeuralUCB configuration on quantum routing efficiency and stability under stochastic and adversarial conditions. That track record—using existing path-selection and forecasting ideas as building blocks rather than reinventing them—is exactly why it makes sense to keep borrowing from newer, stronger path-exploration and EXA-GP work in the next phase, by plugging their routing policies in as alternative learners and potentially replacing the NeuralUCB component when interpretability or transferability is the priority.

Best,
Piter

On Mon, Dec 1, 2025 at 7:21AM Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hey Dan,

My apologies for the delay. I meant to send this earlier. Here are two quick answers I put together right after our meeting. I can elaborate further in our meeting today if necessary.

1) What is the specific quantum network challenge related to path selection?

The main challenge is to dynamically select multi-hop entanglement routes that maintain high end-to-end fidelity in noisy, lossy, and resource-limited quantum networks. At the same time, the routing strategy must adapt online to changing channel conditions and potential attacks while balancing fidelity, purification overhead, and coherence lifetimes.

2) Could the ML aspect be designed to solve a unique quantum problem?

Yes. The ML component is designed to learn routing policies that map quantum-specific signals (loss, decoherence, entanglement success rates, topology) to path and capacity decisions, making it a direct solution to the quantum routing problem rather than a generic traffic optimization, since it must internalize quantum-mechanical constraints and tradeoffs.

Best,
Piter

On Tue, Nov 25, 2025 at 10:42AM Daniel Krutz <dxkvse@rit.edu> wrote:

Answer each of these in 2-3 sentences. Be right to the point:

- What is the specific quantum network challenge related to path selection?
- Could the ML aspect be designed to solve a unique quantum problem?

On Nov 25, 2025, at 6:10AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Dan,

Thanks for sharing the EXA-GP proposal. I've scanned it briefly and wanted to provide my thoughts on how it fits with our current work.

From a conceptual and architectural standpoint, the proposed EXA-GP direction is not a pivot but a natural, methodologically consistent extension of the framework we have already constructed. Across the literature we reviewed and the internal analyses we performed, three limitations emerge as structurally constraining for current quantum routing techniques: a lack of interpretability, poor transferability across topologies, and single-objective reward structures that obscure competing tradeoffs. Our empirical work with context-aware neural bandits—particularly iCPursuitNeuralUCB—has demonstrated strong performance under stochastic and partially structured

environments, yet these models inherently encode their decision logic within opaque neural activations, constraining explainability and scalability. Evolutionary graph-based genetic programming directly addresses these gaps: evolved policies operate as explicit condition–action structures, scale more naturally across network configurations, and support multi-objective fitness optimization. Importantly, this is not presented as a replacement for our existing models but as a complementary decision layer. The modularity of our architecture (Configuration → Evaluator → Models → Visualizer) was designed precisely to allow heterogeneous policy learners to coexist, making EXA-GP a structurally aligned addition that enables direct, controlled comparison between interpretability-driven and adaptivity-driven routing strategies.

Methodologically, integrating EXA-GP fits squarely within the phased research trajectory outlined in our GA notes and in the related work synthesis, comparative baselines, and stochastic-first evaluation. Beginning with stochastic benchmarking allows us to quantify variability and establish rigorous baselines before layering in more complex threat dynamics. This scaffolding mirrors my research plan: each module—predictive forecasting, allocator adaptation, evolutionary rule formation—contributes distinct forms of intelligence, but their interaction yields a more robust and generalizable decision system, aka the puzzle plan. EXA-GP strengthens the framework by providing interpretable, transferable policy structures that neural bandits can adapt around, especially in settings where regulatory transparency or cross-topology deployment are required. In the context of the work I have done, this next-level direction is not an incremental extension of our current paper but a forward-looking research pathway that tests whether evolutionary–bandit hybrids can overcome the precise limitations the literature identifies. It positions our future work at a point where methodological innovation, architectural extensibility, and my research framework's systems-level coherence meet—ensuring alignment with both our academic goals and the expectations guiding the project.

Let me know if you'd like me to prepare any additional materials or clarifications.

Best,
Piter